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# What does self-supervision add to engineered features for stellar light curves?

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## Abstract

1 Self-supervised models promise one representation, pre-trained once on unlabelled  
2 survey data and read out cheaply for many downstream tasks. In stellar time-  
3 domain astronomy that promise has to improve on a strong, cheap incumbent: a  
4 few dozen engineered light-curve statistics. We pre-train a Conv1D variational  
5 encoder with a recurrent latent-dynamics model on 195,883 unlabelled TESS 2-  
6 minute light curves, freeze it, and ask whether the features it learns add anything  
7 to 25 engineered features under one linear readout, on ten labelled tasks spanning  
8 variability classification, asteroseismology and rotation. Concatenating the two  
9 improves the readout beyond 2 SE on 7 of 10 tasks; an untrained encoder does  
10 not. Against two networks trained end-to-end on the same labelled stars, splits  
11 and inputs, the frozen readout is ahead of a supervised Conv1D with the encoder's  
12 architecture on 6 of 10 tasks and indistinguishable on the other 4, and ahead of  
13 an MLP on raw flux on 9 of 10. **Self-supervision adds to engineered features  
14 rather than replacing them. Under a plain logistic regression, the combination  
15 matches or exceeds the two supervised networks we train on the same 669 to  
16 16,002 labelled stars per task.**

## 17 1 Introduction

18 Wide-field photometric surveys now produce light curves far faster than they can be labelled. TESS  
19 [Ricker et al., 2015] reads out hundreds of thousands of pre-selected stars at 2-minute cadence, its  
20 full-frame images, now every 200 s, cover every star in the field, and LSST [Ivezić et al., 2019] will  
21 enlarge the gap by orders of magnitude. Self-supervised learning is the natural response: pre-train  
22 one encoder on the unlabelled archive, then attach a cheap readout for each scientific question. This  
23 shape has recently been pursued for astronomical light curves by Donoso-Oliva et al. [2025], Zuo  
24 et al. [2025] and Rui [2026], and for TESS by Poznanski [2026], who validates with embedding  
25 diagnostics rather than downstream tasks.

26 What makes this domain unusual is the strength of the incumbent it must improve on. Decades of  
27 variable-star research have distilled light curves into compact engineered summaries – periodogram  
28 peak structure, band-limited power, spectral entropy, robust scatter statistics – and pipelines built on  
29 them remain highly competitive [Audenaert et al., 2021]. The benchmark of Li et al. [2025] makes  
30 the point sharply: across seven expert-labelled variable-star classes, no time-series foundation model  
31 they tested strictly surpassed the long-established domain feature baseline on classification. A learned  
32 representation for this data therefore has to answer a specific question, not a generic one: does it  
33 carry information the engineered features do not already have?

34 We take that question literally and measure it. We pre-train a latent-dynamics encoder on unlabelled  
35 TESS photometry, freeze it, and ask what the features it learns add to a 25-feature engineered stack  
36 under an identical linear readout – and whether the resulting pipeline, which trains no deep network

at readout time, matches or exceeds two networks trained end-to-end on the same labels. Our contributions are:

- **The learned features are complementary to the engineered ones, not a replacement for them.** Concatenating the two improves a frozen linear readout beyond 2 SE on 7 of 10 tasks, with the gains concentrated where the engineered basis is information-starved; an untrained-encoder control shows the effect is not generic column addition.
- **The combined pipeline, read out by logistic regression, matches or exceeds the two supervised networks we train end-to-end on the same labels.** Against a supervised Conv1D with the encoder’s architecture it is ahead on 6 of 10 tasks and indistinguishable on the other 4; against an MLP on raw flux it is ahead on 9 of 10.

## 2 Data and method

**Data.** We use the 2-minute SPOC PDCSAP target light curves of 195,883 stars brighter than  $T = 10$ , one segment per star, drawn from Sectors 1–101; one cadence keeps one window a fixed span of physical time. Each light curve is split into contiguous segments at runs of missing cadences and at time gaps exceeding five times the median cadence; segments are median-normalised by their median absolute deviation and sliced into non-overlapping windows of 256 cadences ( $256 \times 120 \text{ s} \approx 8.5 \text{ h}$ ). Missing values are never interpolated or padded – any window containing one is discarded, and segments are never stitched across sector or gap boundaries. Pre-training consumes sequences of 16 consecutive windows ( $\approx 5.69 \text{ d}$ ) and uses no labels of any kind.

**Encoder.** A 1D convolutional variational encoder [Kingma and Welling, 2014] maps each window to a 128-dimensional Gaussian posterior, and a GRU [Cho et al., 2014] predicts the next window’s latent from its predecessors, run both forward and backward over the sequence. The training objective combines window reconstruction, the KL term, the latent-dynamics prediction error, and a Hann-tapered spectral auxiliary term that penalises reconstruction error in the log periodogram. The encoder is trained once with six random seeds and then frozen; no gradient from any label ever reaches it.

**Representations and readout.** A star’s learned features are  $\mu$ , the mean over its first segment (16–20 windows) of the per-window posterior means. The engineered baseline is a fixed 25-feature vector modelled on the T’DA variability pipeline [Audenaert et al., 2021]: the three strongest periodogram peaks with their normalised powers and ratios, a harmonic-power ratio, power in eight log-spaced frequency bands, spectral entropy and centroid, skew, kurtosis, point-to-point scatter ratio, and three robust dispersion statistics. Readouts are deterministic and frozen, standardised on the training split alone: class-balanced  $L_2$  logistic regression at a fixed  $C = 1$  for the detection tasks, ridge for the two regressions with its penalty set by leave-one-out cross-validation (cross-validating  $C$  as well moves no reported gain by more than 0.011). We compare four arms, each the same readout on a different feature set: features,  $\mu$ , features  $\oplus \mu$ , and, as a control, the same fusion built on an *untrained* encoder. The two supervised baselines of the next paragraph are the fifth and sixth arms.

**Supervised baselines.** Two end-to-end baselines see exactly the same input windows, star sets and splits. **C1** reuses the encoder’s convolutional trunk verbatim, adds a bottleneck and a scalar head, and is trained per task with a star-level bag loss (the mean of the star’s window logits, taken in-graph) and class weighting. **C2** replaces the convolutional trunk with a dense one and changes nothing else, so **C1** versus **C2** isolates the convolutional inductive bias. Both use one pre-registered regularisation setting for all tasks, select on a validation split no probe has ever touched, and are trained with six seeds.

**Tasks and metrics.** We score ten tasks with one probe per physical quantity: pulsating variables [Gao et al., 2025], eclipsing binaries [Prša et al., 2022], rotation and rotation period [Boyle et al., 2026], transits [Guerrero et al., 2021], flare activity [Seli et al., 2025, Vida et al., 2021], oscillating red giants and their  $\nu_{\text{max}}$  [Hon et al., 2021], solar-like oscillators [Hatt et al., 2023], and RGB versus helium-burning discrimination [Sreenivas et al., 2026]. Here  $\nu_{\text{max}}$  is the frequency of maximum oscillation power. Detection tasks are scored by the area under the precision–recall curve (PR-AUC), whose floor is the positive-class prevalence rather than 0.5; prevalence is therefore quoted with every detection score. The two regressions are scored by  $R^2$ .

Table 1: Ten-task scorecard. Linear-readout arms use a frozen logistic or ridge readout on the mean-pooled representation; the two supervised arms are trained end-to-end per task. Bold marks a leader separated from the runner-up by the combined 2 SE band; italic marks a leader that is only the point-estimate argmax.  $n$  is the test-star count and prev. the positive-class prevalence.

| task              | metric  | $n$ (prev.)  | frozen linear readout |       |                       | end-to-end supervised |              |
|-------------------|---------|--------------|-----------------------|-------|-----------------------|-----------------------|--------------|
|                   |         |              | features              | $\mu$ | features $\oplus \mu$ | Conv1D                | MLP          |
| pulsating         | PR-AUC  | 2021 (0.107) | 0.789                 | 0.806 | <b>0.846</b>          | 0.816                 | 0.748        |
| eclipsing binary  | PR-AUC  | 2021 (0.097) | 0.742                 | 0.771 | <i>0.775</i>          | 0.774                 | 0.719        |
| rotation          | PR-AUC  | 2021 (0.089) | 0.540                 | 0.559 | <i>0.555</i>          | <i>0.565</i>          | 0.526        |
| transit           | PR-AUC  | 2021 (0.060) | 0.190                 | 0.144 | <b>0.218</b>          | 0.134                 | 0.109        |
| oscillating giant | PR-AUC  | 3429 (0.383) | <b>0.920</b>          | 0.854 | 0.918                 | 0.875                 | 0.883        |
| solar-like osc.   | PR-AUC  | 3429 (0.122) | 0.320                 | 0.336 | 0.390                 | <i>0.396</i>          | 0.275        |
| RGB vs. HeB       | ROC-AUC | 161 (0.702)  | 0.758                 | 0.660 | 0.708                 | 0.530                 | <b>0.762</b> |
| $\nu_{\max}$      | $R^2$   | 1313         | 0.831                 | 0.802 | <b>0.867</b>          | 0.832                 | 0.834        |
| rotation period   | $R^2$   | 150          | 0.703                 | 0.677 | <b>0.717</b>          | 0.597                 | 0.698        |
| flare activity    | PR-AUC  | 3429 (0.089) | 0.474                 | 0.453 | 0.524                 | <i>0.533</i>          | 0.444        |

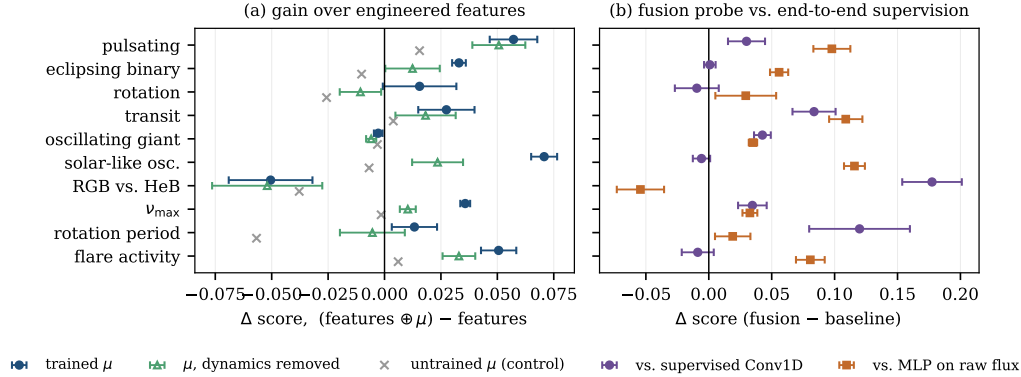


Figure 1: The two claims. **(a)** What the learned features add to the engineered features under an identical frozen linear readout, for the shipped encoder, the same recipe with the latent-dynamics term removed, and an untrained-encoder control. **(b)** The frozen fusion probe against the two end-to-end supervised baselines. Bars are 2 SE; the untrained control is a single encoder and carries none.

**Uncertainties.** All error bars are  $2\times$  the standard error of the mean over seeds. For the  $\mu$  arms the seeds are independent pre-training runs (six); for C1 and C2 they are initialisation and shuffling seeds (six, matching the pre-training count). Because a pre-training seed and an initialisation seed are different objects, differences between a  $\mu$  arm and a supervised arm are treated as unpaired and their bands added in quadrature even at equal counts. The engineered arm is deterministic and carries no seed spread. A difference counts as resolved only when it exceeds the combined 2 SE band. These bands measure re-training spread only; a paired bootstrap over the test stars (1,000 resamples, every arm rescored on the same stars, seeds averaged; Appendix C) excludes zero on 5 of the 7 fusion gains, 5 of the 6 leads over C1 and 7 of the 9 over C2, the rest stay ahead in at least 65% of resamples, none changes sign, and the rows that loosen are the two 150–161-star tasks.

### 3 Results

**The learned features add to the engineered features.** Table 1 and Figure 1a give the comparison. Concatenating  $\mu$  with the 25 features improves the frozen readout beyond 2 SE on 7 of 10 tasks, led by solar-like oscillators  $+0.071$ , pulsating  $+0.057$  and flare activity  $+0.051$ , and reaching  $+0.013$  ( $R^2$ ) on rotation period. Rotation is positive but just short of the bar ( $+0.016$  against 0.016). The two negative tasks are the ones where the engineered basis is already saturated or the probe is small:

oscillating giants ( $-0.003$ , on a baseline of  $0.920$ ) and RGB versus helium-burning ( $-0.051$ , 161 test stars). On both, an untrained-encoder control loses by an amount we cannot resolve from the trained fusion’s – the signature of extra collinear columns diluting a linear fit, not of  $\mu$  carrying misleading content; Appendix A gives both rows in full. Across all ten tasks the untrained control is negative on seven and never exceeds  $+0.016$ , so the gains are specific to what pre-training learned.  $\mu$  on its own does not replace the features: it is ahead on only 4 of 10 tasks alone, which is the same conclusion Li et al. [2025] reach for foundation-model embeddings, and is why we report fusion rather than substitution [Makinen et al., 2024].

**The frozen pipeline matches or exceeds the two supervised networks.** Against C1 – the same convolutional trunk, the same labels, the same input windows, trained end-to-end – the frozen fusion probe is ahead on 6 of 10 tasks and indistinguishable from it on the other 4 (Figure 1b). Against C2, an MLP on raw flux, it is ahead on 9 of 10 and behind on one (RGB versus helium-burning, 161 stars). C1 is not weak: it clears the engineered arm on 5 of 10 and the  $\mu$ -only arm on 4, and on seven tasks the frozen probe recovers  $0.98$ – $1.09$  of its margin above the metric floor. This is a statement about this architecture on 669 to 16,002 labelled stars per task, not about supervision in general.

**The latent-dynamics term is what buys the complementarity.** Replacing the forward-backward dynamics objective with a dynamics-free variant, all else fixed and the same six seeds, reduces the fusion gain on all 10 tasks, and by more than 2 SE on 6 when the two arms are differenced seed by seed (Figure 1a, open triangles; largest losses solar-like oscillators  $-0.047$ , rotation  $-0.026$ ,  $\nu_{\max}$   $-0.025$ ). The prediction term, not the autoencoder alone, supplies what the engineered features lack.

## 4 Discussion

**Model selection must be gated on a held-out probe.** Model selection, not optimisation, is where this pipeline is fragile. The reconstruction objective develops a localised artifact at the window boundary, and none of eleven interventions satisfied both pre-registered acceptance criteria, so the tapered variant shipped on the stated rule, not because it won. Underlying this is one measurement (Appendix B). Across ten pre-training recipes of one sweep, validation reconstruction *anti*-predicts downstream probe score: the Spearman  $\rho$  between negated validation reconstruction and PR-AUC runs from  $-0.58$  to  $-0.84$  over four tasks, the loss-optimal recipe is the dynamics-free one on all four, and trusting the loss costs  $0.021$ – $0.049$  PR-AUC. In the recipe family we ship the sign is mixed ( $\rho$  from  $-0.51$  to  $+0.37$ ): there the loss does not predict the probe, and in neither family does it pick the probe-optimal recipe. Across seeds of one recipe no label-free metric we tried ranks anything. Self-supervised representation work here therefore cannot be tuned on its own objective and must be gated on a held-out probe.

**Limitations and next steps.** The four linear-readout arms (features,  $\mu$ , features  $\oplus \mu$ , and the untrained control) share one linear model. It has few parameters, so it cannot rescue weak features, and it is identical across the four, so the comparison is between feature sets. Our claim is made for these 25 engineered features and a linear readout; we make no claim about other feature sets or more powerful readouts. Our corpus is the bright ( $T < 10$ ) 2-minute sample, and every arm, the two supervised baselines included, sees one 5.7-day segment per star. The fainter, far more numerous stars in the full-frame images, and light curves spanning several sectors (which would let rotation periods longer than 5.7 days be measured), are the obvious next experiments. Our negatives are stars absent from the label catalogues, not stars verified to lack the signal, so a correct detection the catalogue missed counts against us; every detection score is therefore a lower bound. The flare label marks stars that have flared somewhere in their TESS data, not a flare inside the 5.7-day segment we encode, so the task is recognising a flaring star, not finding a flare. When the readout is trained on one percent of the labels, the advantage narrows on 6 of 10 tasks, so how much pre-training helps when labels are scarce remains an open question.

**Reproducibility and disclosure.** §2 gives the corpus cut, geometry, features, objective and readouts; the encoder has four convolutional stages of width  $32/64/128/256$  (kernel 5, each halving length), a 128-dimensional Gaussian latent and a single-layer GRU of width 256. Splits are by star, drawn once and shared by every arm, the supervised baselines included. Code and score tables will be released on publication; generative AI assisted with analysis code and prose editing.

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## 207 A The two tasks where fusion does not help

| task               | features | fusion $\Delta$ | 2 SE  | untrained $\Delta$ | $n_{\text{test}}$ |
|--------------------|----------|-----------------|-------|--------------------|-------------------|
| oscillating giants | 0.920    | -0.003          | 0.002 | -0.003             | 3,429             |
| RGB vs. He-burning | 0.758    | -0.051          | 0.019 | -0.038             | 161               |

209 **Oscillating giants: the engineered basis is already near-sufficient.** The labels of Hon et al. [2021]  
 210 come from a convolutional classifier reading a  $128 \times 128$  image of each star’s power spectrum, which  
 211 fires on a visible oscillation power excess at  $6 \lesssim \nu_{\text{max}} \lesssim 250 \mu\text{Hz}$ . Our engineered vector encodes  
 212 close to that same quantity: eight log-spaced band powers spanning 0.3–360 c/d, plus the spectral  
 213 centroid and entropy, locate the excess and describe its concentration. The features accordingly reach  
 214 0.920, and the trained and untrained fusion arms then lose the same 0.003 – 128 collinear columns  
 215 diluting a linear fit, with no trained-versus-untrained difference left to attribute to  $\mu$ .

216 **RGB versus helium-burning: no arm resolves the discriminant.** Sreenivas et al. [2026] assign the  
 217 evolutionary state with a convolutional classifier reading the background-corrected power spectrum  
 218 *folded at  $\Delta\nu$* ; the physics underneath is the dipole mixed-mode period spacing  $\Delta\Pi_1$  [Bedding  
 219 et al., 2011]. Neither is accessible at our baseline. The encoder sees one 256-cadence window at a  
 220 time, so the finest frequency structure available to  $\mu$  is  $\Delta f = 32.6 \mu\text{Hz}$ , while  $\Delta\nu$  for these stars  
 221 is  $\approx 1.5\text{--}18 \mu\text{Hz}$  – the comb itself is unresolved, let alone a fold of it. The engineered features,  
 222 computed on the whole 5.69 d segment, reach  $2.0 \mu\text{Hz}$ , still coarser than the mixed-mode spacing  
 223  $\Delta\Pi_1\nu^2 \approx 0.2\text{--}0.6 \mu\text{Hz}$  near  $\nu_{\text{max}} = 50 \mu\text{Hz}$ . No arm on this row measures the quantity the label  
 224 encodes; the 0.758 the features reach rides on correlates of evolutionary state rather than on its  
 225 discriminant. The row behaves accordingly: it is the only one where C2 (0.762) is ahead of the frozen  
 226 probe while C1 reaches only 0.530 on 755 training stars, and it carries the smallest test set in the  
 227 study.

## 228 B Validation reconstruction against downstream score

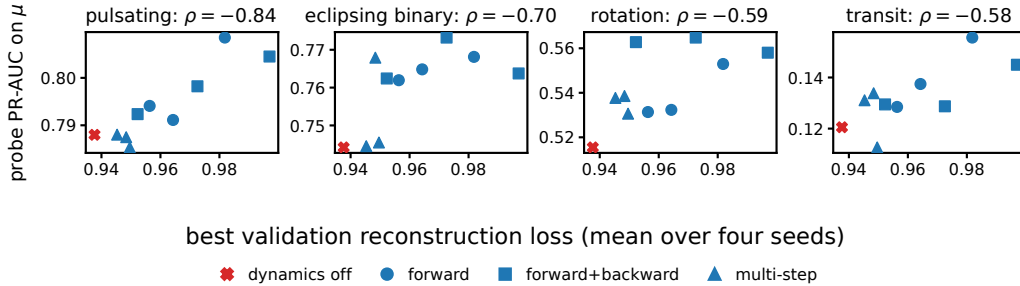


Figure 2: One point per pre-training recipe (mean over its four seeds), one panel per task. The ten recipes share corpus, architecture and loss terms and differ only in the latent-dynamics term (off, forward, forward+backward, multi-step, the last three at three weights).  $x$ : best post-warm-up validation reconstruction loss;  $y$ : PR-AUC of the frozen linear probe on  $\mu$ ;  $\rho$ : Spearman’s between negated loss and score.

Figure 2 is the measurement behind the model-selection paragraph of §4. The loss-optimal recipe is the dynamics-free one (red cross) on all four tasks, and it is the worst or next-to-worst probe on each, so selecting on the loss costs 0.021–0.049 PR-AUC against the probe-optimal recipe. The trend is not carried by that one point: with it removed,  $\rho$  still runs from  $-0.43$  to  $-0.87$ . The recipes are an earlier sweep than the shipped encoder (they carry an earlier spectral auxiliary term), so we repeat the estimator on the family we ship: 19 recipes built on the final encoder that vary the form, weight and floor of the auxiliary term, six seeds each (two more recipes in which one seed collapsed are excluded; we never drop a seed). There the anti-correlation does not persist as such:  $\rho$  is  $-0.51$  on pulsating and  $-0.50$  on transit,  $+0.37$  on eclipsing binaries and  $+0.11$  on rotation. What persists is that the loss is not a selector: its optimum is again the recipe with the weakest competing term (no auxiliary term at all), which is never the probe optimum, and trusting it costs 0.011–0.034 PR-AUC. So the figure shows an anti-correlation on every task in one family; in the shipped family it survives on two tasks and reverses on two; in neither does the loss choose. Both are statements across recipes only: within one recipe, across its seeds, no label-free curve metric we tried (reconstruction, reconstruction plus auxiliary, active latent units) ranks the seeds, every such correlation sitting within  $\pm 0.3$  of zero.

## C Paired bootstrap over test stars

The 2 SE bands in the body come from re-training and say nothing about whether the particular test stars favoured one arm. For every printed contrast we therefore resample each task’s test set with replacement 1,000 times, rescore every arm on the identical resampled list from its saved per-star predictions (no model is retrained), average each arm over its seeds, and take the difference. Table 2 gives the 95% percentile interval of that difference and the share of resamples in which the fusion arm is ahead; intervals that exclude zero are bold. The check tightens no verdict and reverses none: every row the body calls ahead has the fusion arm ahead in at least 65% of resamples. It does loosen the smallest tasks – rotation period (150 test stars) and RGB versus helium-burning (161) account for four of the ten rows where a resolved 2 SE verdict becomes an interval that touches zero.

| task               | fusion – features |          | fusion – C1      |          | fusion – C2      |          | fusion – dynamics-off fusion |          |
|--------------------|-------------------|----------|------------------|----------|------------------|----------|------------------------------|----------|
|                    | 95% CI            | $f_{>0}$ | 95% CI           | $f_{>0}$ | 95% CI           | $f_{>0}$ | 95% CI                       | $f_{>0}$ |
| pulsating          | [+0.012, +0.105]  | 0.99     | [−0.008, +0.073] | 0.94     | [+0.045, +0.150] | 1.00     | [−0.023, +0.036]             | 0.69     |
| eclipsing binary   | [+0.005, +0.062]  | 0.99     | [−0.024, +0.025] | 0.52     | [+0.022, +0.094] | 1.00     | [+0.004, +0.038]             | 0.99     |
| rotation           | [−0.026, +0.057]  | 0.77     | [−0.056, +0.040] | 0.38     | [−0.023, +0.079] | 0.87     | [−0.005, +0.052]             | 0.94     |
| transit            | [−0.025, +0.065]  | 0.85     | [+0.041, +0.124] | 1.00     | [+0.046, +0.174] | 1.00     | [−0.016, +0.035]             | 0.78     |
| oscillating giants | [−0.010, +0.005]  | 0.21     | [+0.028, +0.057] | 1.00     | [+0.021, +0.051] | 1.00     | [−0.002, +0.008]             | 0.87     |
| solar-like osc.    | [+0.044, +0.097]  | 1.00     | [−0.037, +0.022] | 0.34     | [+0.079, +0.151] | 1.00     | [+0.025, +0.067]             | 1.00     |
| RGB vs. HeB        | [−0.087, −0.012]  | 0.00     | [+0.099, +0.247] | 1.00     | [−0.126, +0.020] | 0.08     | [−0.037, +0.039]             | 0.52     |
| $\nu_{\max}$       | [+0.028, +0.043]  | 1.00     | [+0.024, +0.045] | 1.00     | [+0.018, +0.047] | 1.00     | [+0.020, +0.031]             | 1.00     |
| rotation period    | [−0.038, +0.067]  | 0.65     | [+0.054, +0.186] | 1.00     | [−0.053, +0.100] | 0.69     | [−0.021, +0.060]             | 0.83     |
| flare              | [+0.019, +0.078]  | 1.00     | [−0.047, +0.029] | 0.32     | [+0.027, +0.128] | 1.00     | [−0.000, +0.035]             | 0.97     |

Table 2: Paired star-bootstrap of the four contrasts the body counts. Each entry is the 95% interval of (left arm – right arm) over 1,000 resamples of the test stars, seeds averaged, and the share of resamples with the difference above zero ( $f_{>0}$ ; 1.00 means the left arm is ahead in every resample, 0.00 that it is behind in every resample). Bold: interval excludes zero.